

Structural Optimization and Economic Feasibility of an Industrial Robot Arm Bracket

1. Design Load & Operating Scenario

The L-bracket serves as the structural interface between the robot flange and the end effector. To establish the design boundary conditions, a worst case scenario is assumed that $4g$ peak dynamic acceleration is induced as the on a robot arm handling a hypothetical 10 kg payload capacity:

$$F = m \cdot (g + a_{max}) = 10 \text{ kg} \times (9.81 + 40) \text{ m/s}^2 \approx 500 \text{ N}$$

2. Bending Moment at the Root Section

The horizontal leg is evaluated as a cantilever beam with an effective overhang length of $L = 80 \text{ mm}$. The peak bending moment acting on the fixed root section is:

$$M_b = F \cdot L = 500 \text{ N} \cdot 80 \text{ mm} = 40\,000 \text{ N} \cdot \text{mm}$$

3. Allowable Stress & Safety Requirement

Baseline Design A utilizes standard structural steel S235JR ($R_e = 235 \text{ MPa}$). To ensure durability and prevent early plastic deformation under dynamic cyclic loading, an engineering safety factor of $\nu = 2.0$ is defined:

$$\sigma_{bzul} = \frac{\sigma_{bF}}{\nu} = \frac{235 \text{ MPa}}{2.0} = 117.5 \text{ MPa}$$

4. Minimum Wall Thickness Calculation

Considering the bracket width as $B = 80 \text{ mm}$, the required section modulus (W_{erf}) to sustain the bending load is:

$$W_{erf} = \frac{M_b}{\sigma_{bzul}} = \frac{40\,000 \text{ N} \cdot \text{mm}}{117.5 \text{ N/mm}^2} \approx 340.4 \text{ mm}^3$$

Modeling the root section as a rectangular profile ($W = \frac{B \cdot t^2}{6}$), the theoretical minimum wall thickness (t_{erf}) is derived as:

$$W = \frac{B \cdot t^2}{6} \Rightarrow t_{erf} = \sqrt{\frac{6 \cdot W_{erf}}{B}} = \sqrt{\frac{6 \cdot 340.4 \text{ mm}^3}{80 \text{ mm}}} \approx 5.05 \text{ mm}$$

5. Standard Profile Selection

To maintain a safe structural margin, resist dynamic fatigue, and compensate for stress concentrations around the mounting holes, standard profile L 80 × 80 × 8 (DIN EN 10056-1) [2] is selected. The nominal thickness of $t = 8 \text{ mm}$ safely exceeds the analytical minimum requirement of 5.05 mm.

Note: While $\sigma_{bF} \approx 1.2 \cdot R_e$ per Tabellenbuch Metall [1], using unmodified R_e provides a conservative safety margin.

4.4 Stähle, Fertigerzeugnisse

159

Winkelstahl

Gleichschenkliger Winkelstahl, warmgewalzt (Auswahl)

vgl. DIN EN 10056-1 (2017-06)

S Querschnittsfläche

I Flächenmoment 2. Grades

W axiales Widerstandsmoment

m' längenbezogene Masse

Werkstoff: Unlegierter Baustahl DIN EN 10025-2, z. B. S235J0

Lieferart: Von 20 x 20 x 3 bis 250 x 250 x 35, in Herstelllängen: 6 m < 12 m, Festlängen: 6 m < 12 m ± 100 mm

$$r_1 \approx t$$

$$r_2 \approx \frac{t}{2}$$

Kurzzeichen	Abmessungen in mm		S cm ²	m' kg/m	Abstände der Achsen e cm	Für die Biegeachse $x-x$ und $y-y$		Anreißmaße ¹⁾		
L	a	t				$I_x = I_y$ cm ⁴	$W_x = W_y$ cm ³	w_1 mm	w_2 mm	d_1 mm
20 x 20 x 3	20	3	1,12	0,882	0,598	0,39	0,28	12	–	4,3
25 x 25 x 3	25	3	1,42	1,12	0,723	0,80	0,45	15	–	6,4
25 x 25 x 4	25	4	1,85	1,45	0,762	1,02	0,59	15	–	6,5
30 x 30 x 3	30	3	1,74	1,36	0,835	1,40	0,65	17	–	8,4
30 x 30 x 4	30	4	2,27	1,78	0,878	1,80	0,85	17	–	8,4
35 x 35 x 4	35	4	2,67	2,09	1,00	2,95	1,18	18	–	11
40 x 40 x 4	40	4	3,08	2,42	1,12	4,47	1,55	22	–	11
40 x 40 x 5	40	5	3,79	2,97	1,16	5,43	1,91	22	–	11
45 x 45 x 4,5	45	4,5	3,90	3,06	1,25	7,14	2,20	25	–	13
50 x 50 x 4	50	4	3,89	3,06	1,36	8,97	2,46	30	–	13
50 x 50 x 5	50	5	4,80	3,77	1,40	11,0	3,05	30	–	13
50 x 50 x 6	50	6	5,69	4,47	1,45	12,8	3,61	30	–	13
60 x 60 x 5	60	5	5,82	4,57	1,64	19,4	4,45	35	–	17
60 x 60 x 6	60	6	6,91	5,42	1,69	22,8	5,29	35	–	17
60 x 60 x 8	60	8	9,03	7,09	1,77	29,2	6,89	35	–	17
65 x 65 x 7	65	7	8,70	6,83	1,85	33,4	7,18	35	–	21
70 x 70 x 6	70	6	8,13	6,38	1,93	36,9	7,27	40	–	21
70 x 70 x 7	70	7	9,40	7,38	1,97	42,3	8,41	40	–	21
75 x 75 x 6	75	6	8,73	6,85	2,05	45,8	8,41	40	–	23
75 x 75 x 8	75	8	11,4	8,99	2,14	59,1	11,0	40	–	23
80 x 80 x 8	80	8	12,3	9,63	2,26	72,2	12,6	45	–	23
80 x 80 x 10	80	10	15,1	11,9	2,34	87,5	15,4	45	–	23
90 x 90 x 7	90	7	12,2	9,61	2,45	92,6	14,1	50	–	25
90 x 90 x 8	90	8	13,9	10,9	2,50	104	16,1	50	–	25

Figure 5.1: Geometric dimensions for equal leg angle profiles per DIN EN 10056-1 [2].

6. Fastener and Hole Selection

To secure the bracket assembly under static and dynamic loading, standard metric bolts and appropriate clearance dimensions must be established. The selection must comply with the geometric constraints of the chosen profile and prevent any local surface yield.

6.1. Fastener Strength Class

An M16 bolt is selected to carry the applied $F = 500$ N load and provide sufficient clamping force. To ensure high fatigue resistance and prevent plastic deformation under dynamic operating conditions, Property Class 8.8 per ISO 898-1 [3] is specified.

Festigkeitsklassen und Werkstoffkennwerte											
 	Werkstoffkennwerte		Festigkeitsklassen für Schrauben aus								
			unlegierten und legierten Stählen					nichtrostenden Stählen ¹⁾			
			5.8	6.8	8.8	9.8	10.9	12.9	A2-50	A4-50	A2-70
	Zugfestigkeit R_m in N/mm ²		500	600	800	900	1000	1200	500	500	700
	Streckgrenze R_e in N/mm ²		400	480	640	720	900	1080	210	210	450
	Bruchdehnung A in %		–	–	12	10	9	8	20	20	13
1) Die Werkstoffkennwerte gelten für Gewinde ≤ M20.											

¹⁾ Die Werkstoffkennwerte gelten für Gewinde $\leq$ M20.

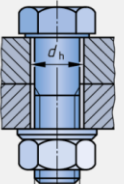
Figure 6.1: Mechanical properties of bolt strength classes per ISO 898-1 [3].

As detailed in Figure 6.1, a class 8.8 fastener provides a nominal tensile strength of $R_m = 800$ MPa and a yield strength of $R_e = 640$ MPa, offering a massive safety margin for this application.

6.2. Clearance Hole and Profile Constraints

According to the geometric properties of the $L 80 \times 80 \times 8$ standard profile (Figure 5.1), the maximum allowable hole diameter for the flange is structurally limited to $d_1 = 23$ mm.

To allow easy assembly while preventing loose play, the through hole is selected per DIN EN 20273 [4].

Durchgangslöcher für Schrauben										vgl. DIN EN 20273 (1992-02)					
	Ge- winde <i>d</i>	Durchgangsloch <i>d_h</i> ¹⁾ Reihe			Ge- winde <i>d</i>	Durchgangsloch <i>d_h</i> ¹⁾ Reihe			Ge- winde <i>d</i>	Durchgangsloch <i>d_h</i> ¹⁾ Reihe					
		fein	mittel	grob		fein	mittel	grob		fein	mittel	grob			
	M1	1,1	1,2	1,3	M5	5,3	5,5	5,8	M24	25	26	28			
	M1,2	1,3	1,4	1,5	M6	6,4	6,6	7	M30	31	33	35			
	M1,6	1,7	1,8	2	M8	8,4	9	10	M36	37	39	42			
	M2	2,2	2,4	2,6	M10	10,5	11	12	M42	43	45	48			
	M2,5	2,7	2,9	3,1	M12	13	13,5	14,5	M48	50	52	56			
	M3	3,2	3,4	3,6	M16	17	17,5	18,5	M56	58	62	66			
	M4	4,3	4,5	4,8	M20	21	22	24	M64	66	70	74			

¹⁾ Toleranzklassen für *d_h*: Reihe fein: H12, Reihe mittel: H13, Reihe grob: H14

¹⁾ Toleranzklassen für d_h : Reihe fein: H12, Reihe mittel: H13, Reihe grob: H14

Figure 6.2: Clearance hole dimensions for metric fasteners per DIN EN 20273 [4].

As shown in Figure 6.2, the medium tolerance series specifies a nominal clearance hole of $d_h = 17.5$ mm. This selection satisfies the profile's geometric constraint ($17.5 \text{ mm} \leq 23.0 \text{ mm}$) while ensuring sufficient clamping surface.

6.3. Washer Selection

Since the bracket is made of aluminum (EN AW-6061-T6), high contact pressure can cause local surface indentation. A flat washer is added to distribute the clamping force and protect the softer material.

A standard chamfered flat washer per DIN EN ISO 7090 (Normal Series) [5] is selected.

Flache Scheiben mit Fase, normale Reihe

vgl. DIN EN ISO 7090 (2000-11)

Härteklasse 200 HV geeignet für:

- Sechskantschrauben und -mutter mit Festigkeitsklassen ≤ 8.8 bzw. ≤ 8 (Mutter)
- Sechskantschrauben und -mutter aus nichtrostendem Stahl

Härteklasse 300 HV geeignet für:

- Sechskantschrauben und -mutter mit Festigkeitsklassen ≤ 10.9 bzw. ≤ 10 (Mutter)

für Gewinde	M5	M6	M8	M10	M12	M16	M20
Nenngröße	5	6	8	10	12	16	20
d_1 min. ¹⁾	5,3	6,4	8,4	10,5	13,0	17,0	21,0
d_2 max. ¹⁾	10,0	12,0	16,0	20,0	24,0	30,0	37,0
h ¹⁾	1	1,6	1,6	2	2,5	3	3
für Gewinde	M24	M30	M36	M42	M48	M56	M64
Nenngröße	24	30	36	42	48	56	64
d_1 min. ¹⁾	25,0	31,0	37,0	45,0	52,0	62,0	70,0
d_2 max. ¹⁾	44,0	56,0	66,0	78,0	92,0	105,0	115,0
h ¹⁾	4	4	5	8	8	10	10
Werkstoffe ²⁾	Stahl			Nichtrostender Stahl			
Sorte	–		–		A2, A4, F1, C1, C4 (ISO 3506) ³⁾		
Härteklasse	200 HV		300 HV (vergütet)		200 HV		
⇒	Scheibe ISO 7090-20-200 HV: Nenngröße (= Gewinde-Nenn- $\varnothing$) = 20 mm, Härteklasse 200 HV, aus Stahl						

¹⁾ jeweils Nennmaße
²⁾ Nichtisenmetalle und andere Werkstoffe nach Vereinbarung
³⁾ vgl. Seite 220

Figure 6.3: Standard dimensions for normal series flat washers per DIN EN ISO 7090 [5].

Based on Figure 6.3, the required washer for an M16 thread has an inner diameter of $d_1 = 17.0$ mm, an outer diameter of $d_2 = 30.0$ mm, and a thickness of $h = 3.0$ mm. To accommodate this 30 mm washer and standard socket tool clearances, a planar mounting boss with a diameter of $D_{\text{boss}} = 34$ mm is implemented in the CAD geometry.

7. Baseline CAD Modeling and Technical Drawing

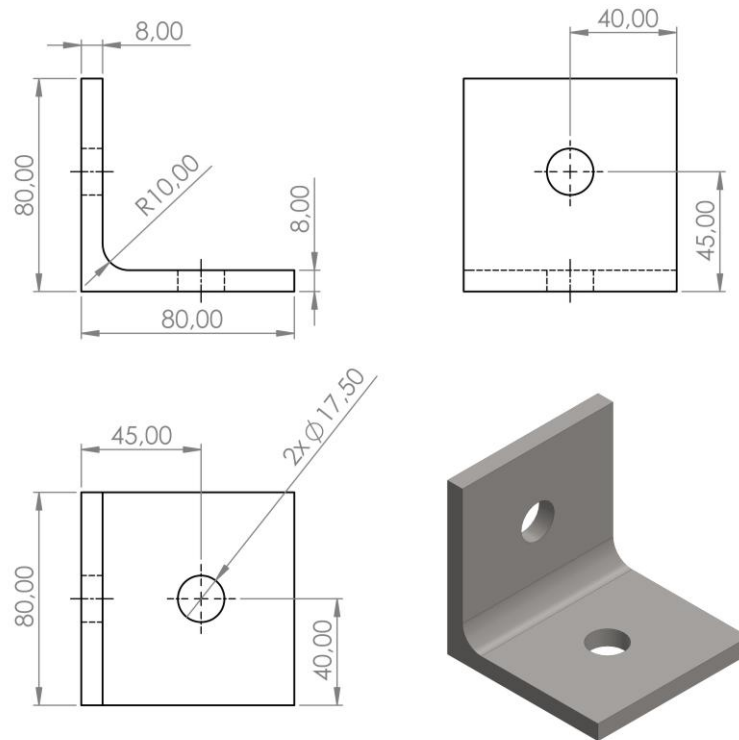


Figure 7.1: Technical drawing and nominal dimensions of the baseline steel L bracket (S235JR).

The baseline CAD model of the L bracket has dimensions of $80 \times 80 \times 80$ mm, a uniform thickness of $t = 8.0$ mm, and an inner fillet radius of $R = 10.0$ mm. To accommodate the M16 bolts and washers selected in Section 6, two clearance holes with a diameter of $\varnothing 17.5$ mm are positioned at the center of each face.

The complete technical drawing is provided in Appendix A. This geometry serves as the reference model for the structural analysis.

8. Structural Analysis of the Baseline Bracket

8.1. Material Properties

The baseline bracket is manufactured from structural steel S235JR (material number 1.0037). In ANSYS Mechanical, the material is modeled as linear elastic and isotropic. The yield strength of 235 MPa is defined as the critical threshold for the structural safety factor evaluation based on the von Mises equivalent stress criterion.

Table 8.1: Mechanical properties of S235JR steel defined in ANSYS Engineering Data.

Property	Symbol	Value	Unit
Density	ρ	7850	kg/m ³
Young's Modulus	E	2.1×10^5	MPa
Poisson's Ratio	ν	0.30	—
Bulk Modulus	K	1.75×10^{11}	Pa
Shear Modulus	G	8.077×10^{10}	Pa
Tensile Yield Strength	R_{eH}	235	MPa

8.2 Fatigue Properties (Synthetic S-N Curve)

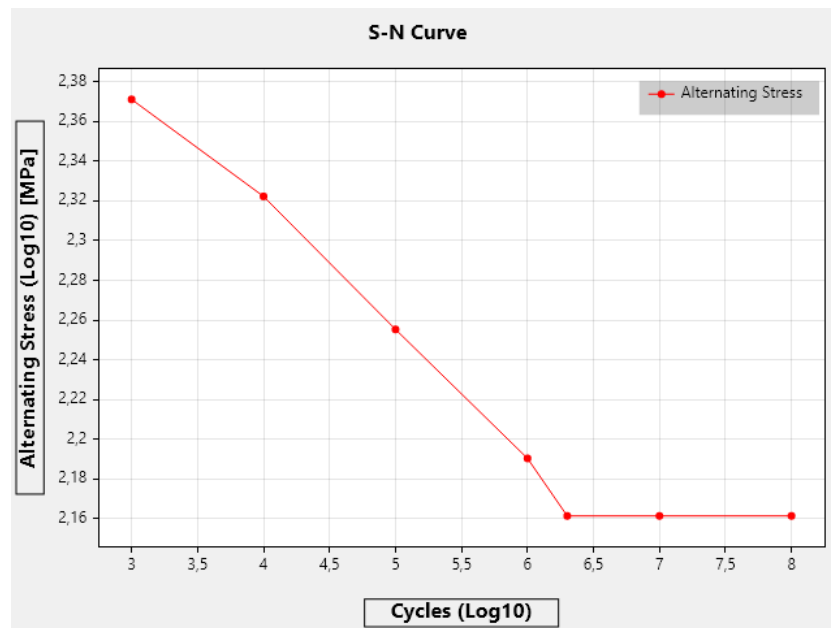


Figure 8.2: Synthetic S-N curve for S235JR steel.

Due to the absence of experimental fatigue test data, a synthetic S-N curve for S235JR was defined according to standard analytical guidelines for fully reversed loading ($R = -1$). The endurance limit was set to 145 MPa ($\approx 0.4 \cdot R_m$) with an elastic threshold at the yield strength (235 MPa).

8.3. Boundary Conditions and Applied Load

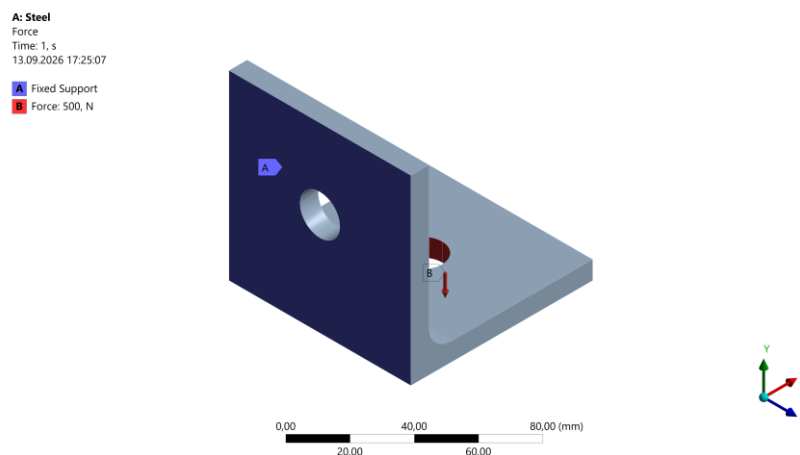


Figure 8.3: Boundary conditions and applied load on the baseline bracket model.

Boundary conditions were applied in ANSYS Mechanical as shown in Figure 8.2. The rear face of the vertical section was fully constrained using a fixed support. A static downward load of 500 N was applied to the inner surface of the hole on the horizontal section in the negative Y direction

Table 8.3: Mesh convergence results for the baseline bracket model.

Solution Number	Nodes	Elements	Max. Equivalent Stress [MPa]	Relative Change [%]
1	1,827	825	14.59	—
2	7,254	4,091	18.93	25.89
3	25,924	16,681	20.44	7.67
4	106,531	73,290	20.67	1.13

8.4. Mesh Convergence Study

A mesh convergence study was conducted using adaptive refinement on the maximum equivalent stress to ensure mesh independent results. The allowable change criterion was set to 2.5%. Convergence was achieved at the fourth iteration with 73,290 elements, where the relative change dropped to 1.13%, yielding a maximum equivalent stress of 20.67 MPa.

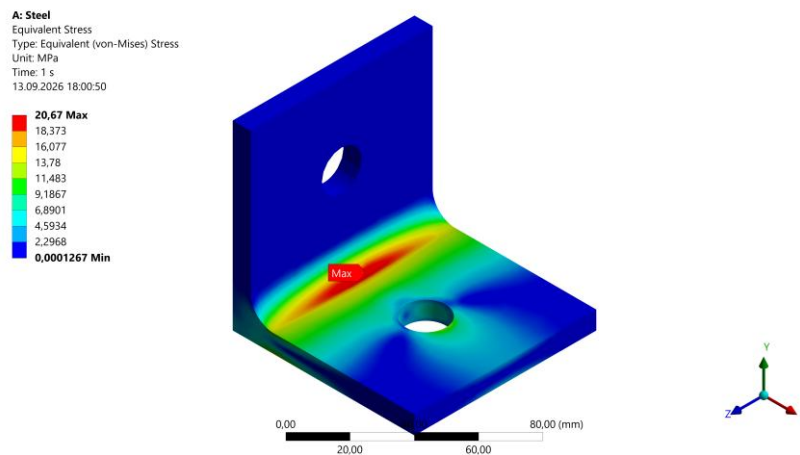


Figure 8.5: Equivalent (von Mises) stress distribution on the baseline bracket model.

8.5. Static Stress Results

The FEA yields a converged maximum equivalent stress of 20.67 MPa under the 500 N peak load, as shown in Figure 8.3. The peak stress is concentrated along the inner radius between the vertical and horizontal sections due to bending. Comparing this result with the yield strength of S235JR steel (235 MPa) confirms that the structure remains well within the elastic region with a static safety factor above 11.

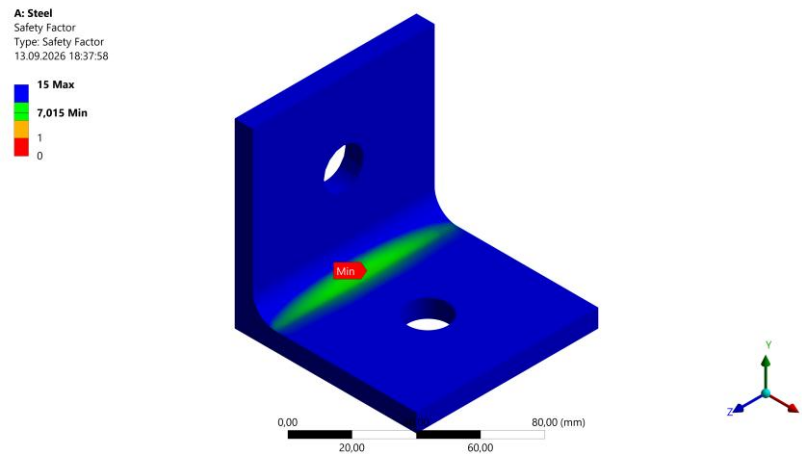


Figure 8.6: Fatigue safety factor distribution of the bracket model.

8.6. Fatigue Analysis Results

A fatigue assessment was conducted under fully reversed cyclic loading using the stress life method as shown in Figure 8.4. The minimum fatigue safety factor was determined as 7.015 at the inner radius between the vertical and horizontal sections. As a consequence of this low stress amplitude, the entire structure achieves infinite fatigue life exceeding 10^8 cycles.

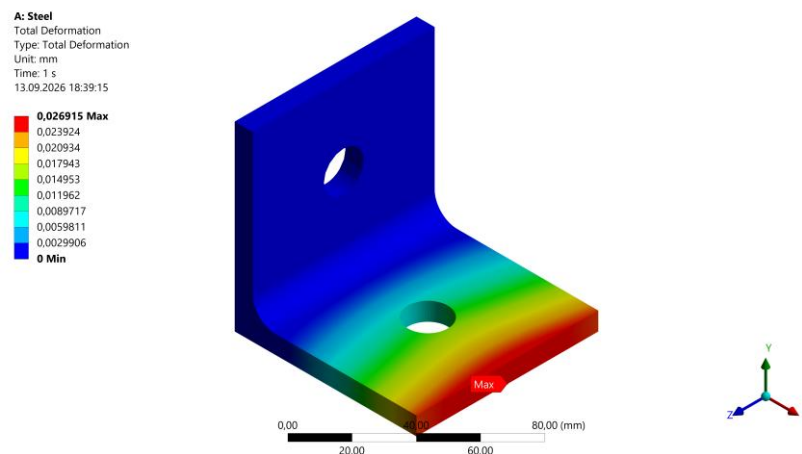


Figure 8.7: Total deformation distribution of the baseline bracket model.

8.7. Deformation Results

The total displacement distribution resulting from the 500 N peak load is presented in Figure 8.5. The maximum deflection is 0.027 mm, located at the free outer edge of the horizontal flange. Considering the 80 mm total cantilever length from the fixed support, this deformation corresponds to approximately $L/3000$, well within the standard industrial machine stiffness limit of $L/1000$ (0.080 mm). This confirms high structural rigidity and ensures that elastic deflections will not impair functional clearances.

8.8. Structural Evaluation and Lightweighting Strategy

The baseline structural steel (S235JR) bracket is substantially over dimensioned under the operational 500 N peak load, exhibiting a static safety factor exceeding 11, a minimum fatigue safety factor of 7.015, and a peak deflection of only 0.027 mm. Given these excessive safety margins, the bracket does not require the mechanical strength offered by structural steel. Consequently, the material will be substituted with a lower strength alternative, EN AW-6061-T6 aluminum alloy, to evaluate structural adequacy in the subsequent analyses.

9. Structural Analysis of the EN AW-6061-T6 Bracket

9.1. Material Properties

The bracket material is substituted with precipitation-hardened aluminum alloy EN AW-6061-T6 (material number 3.3211 / AlMg1SiCu). In ANSYS Mechanical, the material is modeled as linear elastic and isotropic. A yield strength of 275 MPa is defined as the critical threshold for the structural safety factor evaluation based on the von Mises equivalent stress criterion.

Table 9.1: Mechanical properties of EN AW-6061-T6 aluminum alloy defined in ANSYS Engineering Data.

Property	Symbol	Value	Unit
Density	ρ	2700	kg/m ³
Young's Modulus	E	6.9×10^4	MPa
Poisson's Ratio	ν	0.33	—
Bulk Modulus	K	6.765×10^{10}	Pa
Shear Modulus	G	2.594×10^{10}	Pa
Tensile Yield Strength	$R_{p0.2}$	275	MPa

9.2. Fatigue Properties (Synthetic S-N Curve)

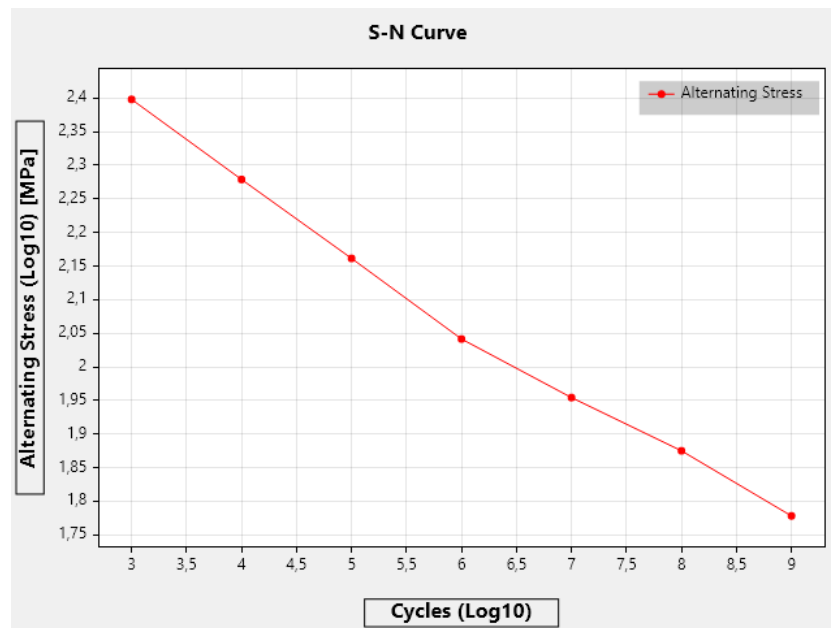


Figure 9.2: S-N curve for EN AW-6061-T6 aluminum alloy.

The fatigue behavior of EN AW-6061-T6 is defined using the S-N curve for fully reversed loading ($R = -1$). In accordance with typical non ferrous material characteristics, the fatigue strength decreases continuously across the cycle range without exhibiting a distinct endurance limit.

9.3. Boundary Conditions and Applied Load

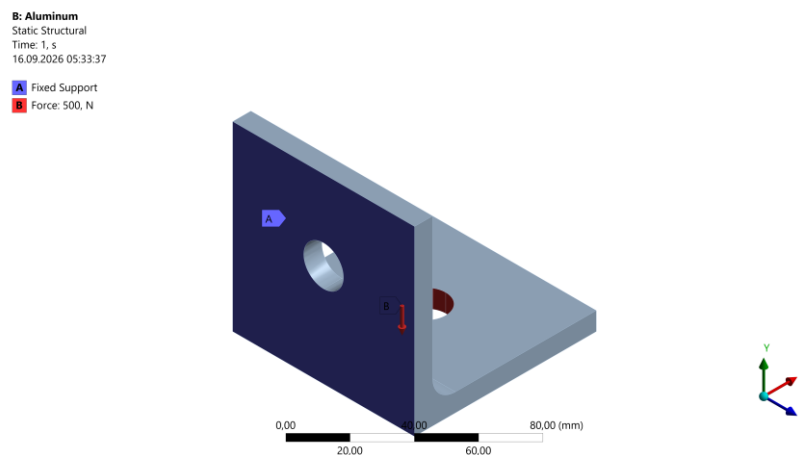


Figure 9.3: Boundary conditions and applied load for the EN AW-6061-T6 bracket model.

The boundary conditions and load application remained identical to the baseline configuration, as illustrated in Figure 9.3. The rear surface of the vertical section was fully constrained using a fixed support, and a static downward load of 500 N was applied to the inner cylindrical surface of the horizontal hole in the negative Y direction.

Table 9.4: Mesh convergence results for the **EN AW-6061-T6** bracket model.

Solution Number	Nodes	Elements	Max. Equivalent Stress [MPa]	Relative Change [%]
1	1,827	825	14.52	—
2	7,254	4,091	18.89	26.17
3	25,783	16,577	20.40	7.67
4	106,452	73,226	20.65	1.23

9.4. Mesh Convergence Study

A mesh convergence study was conducted using adaptive refinement on the maximum equivalent stress to ensure mesh independent results for the EN AW-6061-T6 aluminum bracket model. The allowable change criterion was set to 2.5%. Convergence was successfully achieved at the fourth iteration with 73,226 elements (106,452 nodes), where the relative change dropped to 1.23%, well below the convergence threshold. The converged maximum equivalent stress was evaluated as 20.65 MPa.

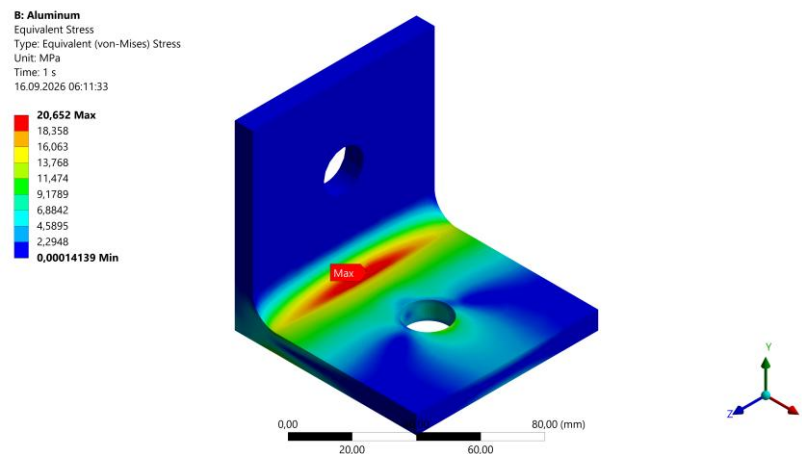


Figure 9.5: Equivalent (von Mises) stress distribution of the EN AW-6061-T6 aluminum bracket model.

9.5. Static Stress Results

The FEA yields a converged maximum equivalent stress of 20.65 MPa under the 500 N peak load, as shown in Figure 9.3. Similar to the baseline steel model, the peak stress is concentrated along the inner fillet radius between the vertical and horizontal sections due to the primary bending moment. Comparing this value with the yield strength of EN AW-6061-T6 aluminum alloy ($R_{p0.2} = 275$ MPa) confirms that the structure operates well within the linear elastic regime, providing a high static safety factor of approximately 13.3 ($S_F = 275/20.65$).

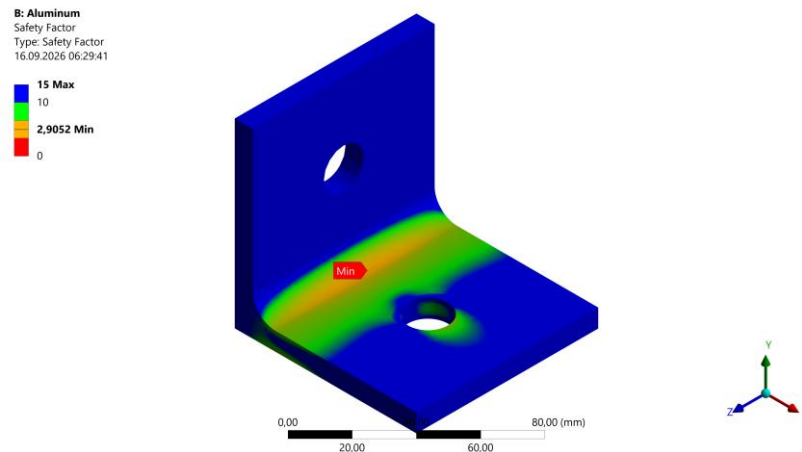


Figure 9.6: Fatigue safety factor distribution of the EN AW-6061-T6 aluminum bracket model.

9.6. Fatigue Analysis Results

A fatigue assessment was conducted under fully reversed cyclic loading using the stress life ($S-N$) approach, as illustrated in Figure 9.4. The minimum fatigue safety factor was determined as 2.91, located along the inner fillet radius between the vertical and horizontal sections where cyclic bending stresses peak.

Because the alternating stress amplitude remains well below the fatigue endurance limit for the specified design criteria, the entire structure achieves infinite fatigue life, reaching the maximum design life limit of 10^9 cycles. Although this safety factor is lower than the baseline steel model (7.02) due to the inherently lower fatigue strength of aluminum alloys, a safety factor of 2.91 remains well above standard industrial design margins, confirming high structural integrity against cyclic failure under nominal 500 N dynamic loading.

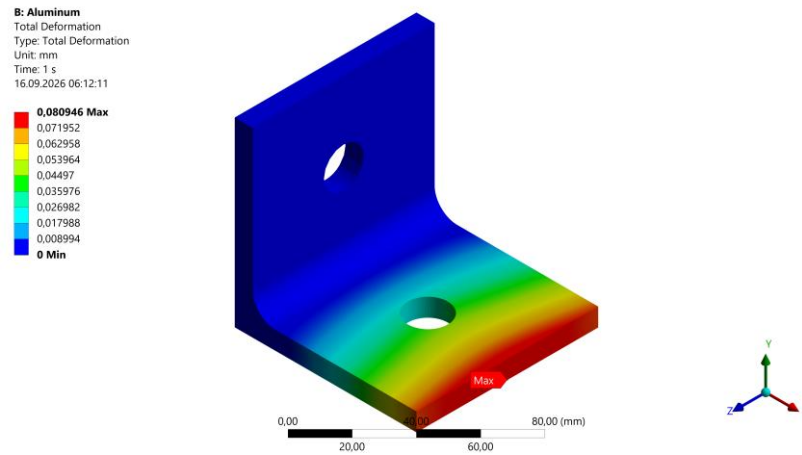


Figure 9.7: Total deformation distribution of the EN AW-6061-T6 aluminum bracket model.

9.7. Deformation Results

The total displacement distribution resulting from the 500 N peak load is presented in Figure 9.7. The maximum deflection is 0.081 mm, located at the free outer edge of the horizontal flange. Because the elastic modulus of aluminum ($E = 69$ GPa) is roughly one third that of structural steel ($E = 210$ GPa), the maximum displacement increases by approximately a factor of 3 compared to the steel baseline (0.027 mm).

Considering the 80 mm cantilever length from the fixed support, this deflection closely aligns with the standard industrial stiffness limit of $L/1000$ (0.080 mm). This verifies that the EN AW-6061-T6 bracket maintains sufficient structural rigidity and functional clearances under peak service loading.

9.8. Structural Evaluation and Lightweighting Strategy

The EN AW-6061-T6 aluminum bracket remains substantially understressed under the operational 500 N peak load, exhibiting a static safety factor of 13.3, a minimum fatigue safety factor of 2.91, and a maximum deflection of 0.081 mm. Although the direct material substitution successfully reduced mass while satisfying the $L/1000$ stiffness criterion, broad areas of the bracket still experience negligible stress levels. Given these remaining design margins, topology optimization will be conducted on the aluminum model to guide geometric lightweighting and evaluate a pocketed design variant in the subsequent analyses.

10. Topology Optimization

10.1. Optimization Setup and Boundary Conditions

To identify inefficient material volumes and guide structural lightweighting, density-based topology optimization was configured on the EN AW-6061-T6 aluminum bracket. The optimization setup and boundary constraints are shown in Figure 10.1.

The bracket body was designated as the active design domain (blue), allowing the solver to redistribute material density based on strain energy distribution. To preserve functional interfaces and structural connectivity, an exclusion region (red) was defined covering the rear fixed mounting face and the internal cylindrical surfaces of both fastener holes. The optimization objective was set to minimize compliance (maximize global stiffness) subject to a response constraint requiring a 50% mass retention of the design space.

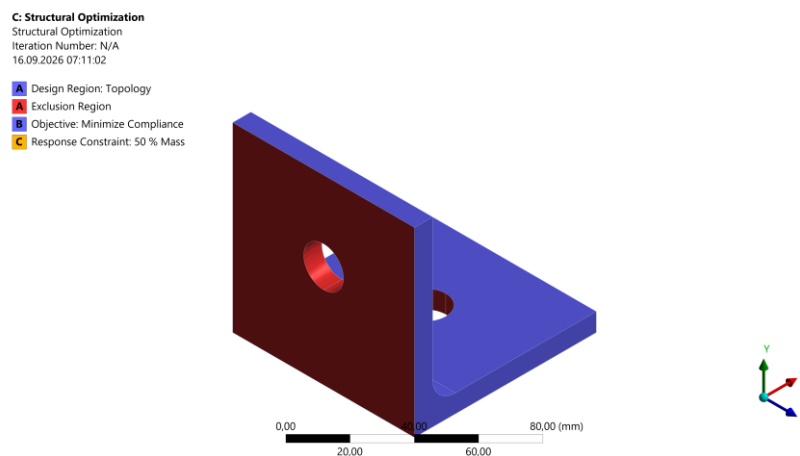


Figure 10.1: Boundary condition setup for topology optimization showing design (blue) and exclusion (red) regions.

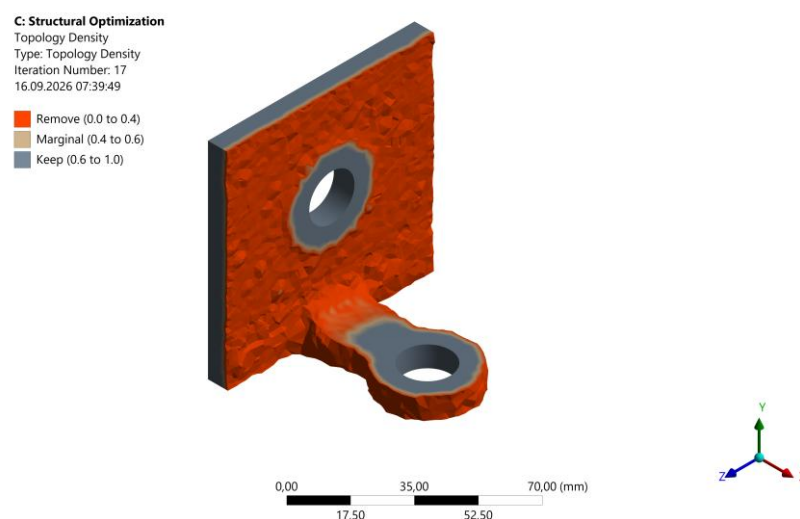


Figure 10.2: Material density distribution obtained from topology optimization.

10.2. Topology Optimization Results

The resulting material density distribution after 17 iterations is presented in Figure 10.2. The optimization algorithm identified a clear, direct load transfer path spanning from the horizontal fastener hole through the central fillet transition to the rear mounting face.

The outer lateral corners of the horizontal flange exhibit near zero strain energy and are identified as redundant material. While the raw organic geometry produced by the solver is unfeasible for conventional machining, it establishes the physical basis for the subsequent CAD redesign. In the redesigned model, the unneeded outer corners are trimmed and contoured around the fastener hole to mirror this primary load path while ensuring conventional manufacturability.

10.3. CAD Redesign Based on Topology Optimization

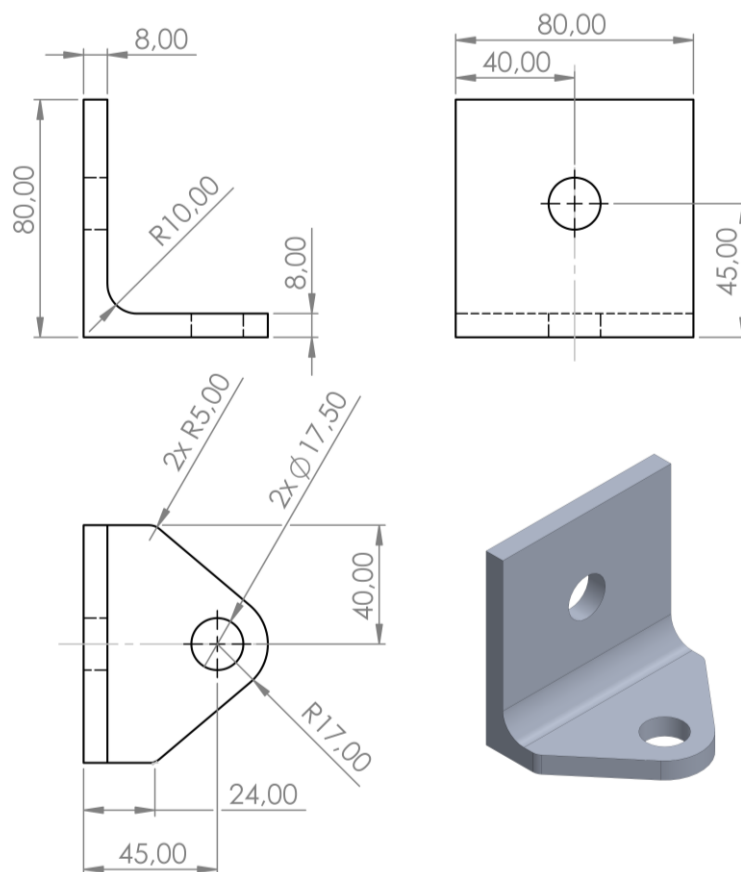


Figure 10.3: Technical manufacturing drawing and 3D CAD model of the optimized lightweight bracket

Based on the topology optimization results, the bracket was redesigned in SolidWorks to remove inactive material. The non load bearing outer corners of the horizontal flange were trimmed, and transition fillets were added to ensure conventional manufacturability while preventing stress concentrations.

10.4 Structural Analysis of the Optimized Bracket

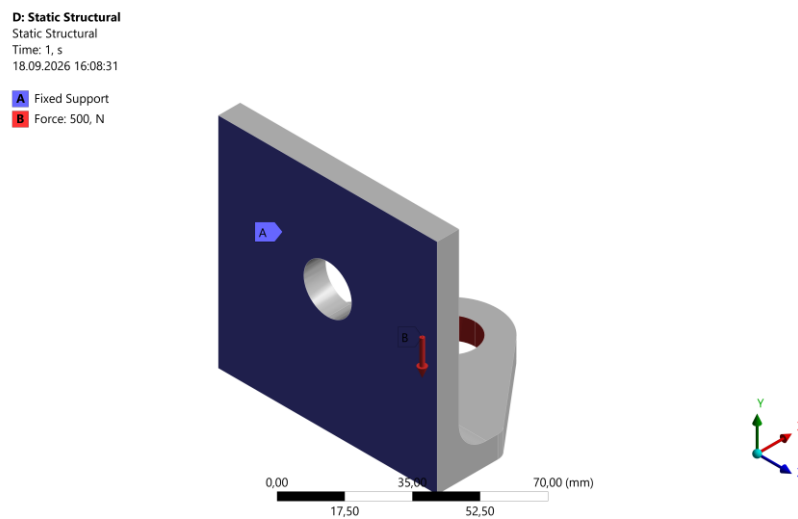


Figure 10.4: Boundary conditions and applied load on the optimized bracket model.

Boundary conditions were applied in ANSYS Mechanical as shown in Figure 10.4. The rear face of the vertical section was fully constrained using a fixed support. A static downward load of 500 N was applied to the inner surface of the hole on the horizontal section in the negative Y direction.

Table 10.1: Mesh convergence results for the optimized bracket model.

Solution Number	Nodes	Elements	Max. Equivalent Stress [MPa]	Relative Change [%]
1	1,866	852	18,39	—
2	8,056	4,619	20,31	9,94
3	30,110	19,646	22,39	9,71
4	104,991	72,500	22,64	1.13

10.5. Mesh Convergence Study

A mesh convergence study was conducted using adaptive refinement on the maximum equivalent stress to ensure mesh-independent results. The allowable change criterion was set to 2.5%. Convergence was achieved at the fourth iteration with 72,500 elements (104,991 nodes), where the relative change dropped to 1.13%, yielding a maximum equivalent stress of 22.64 MPa.

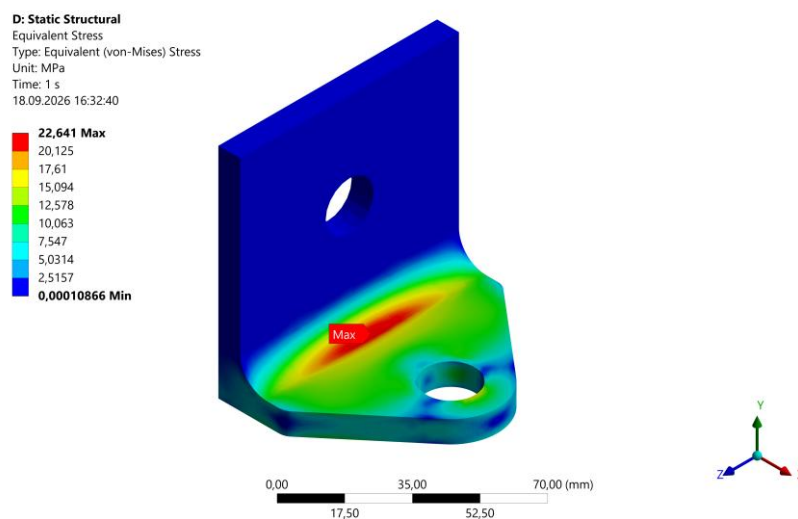


Figure 10.5: Equivalent (von Mises) stress distribution on the optimized bracket model.

10.6. Static Stress Results

The FEA yields a converged maximum equivalent stress of 22.64 MPa under the 500 N peak load, as shown in Figure 10.5. The peak stress is concentrated along the inner radius between the vertical and horizontal sections due to bending. Comparing this result with the yield strength of EN AW-6061-T6 aluminum (240 MPa) confirms that the structure remains well within the elastic region with a static safety factor of approximately 10.6.

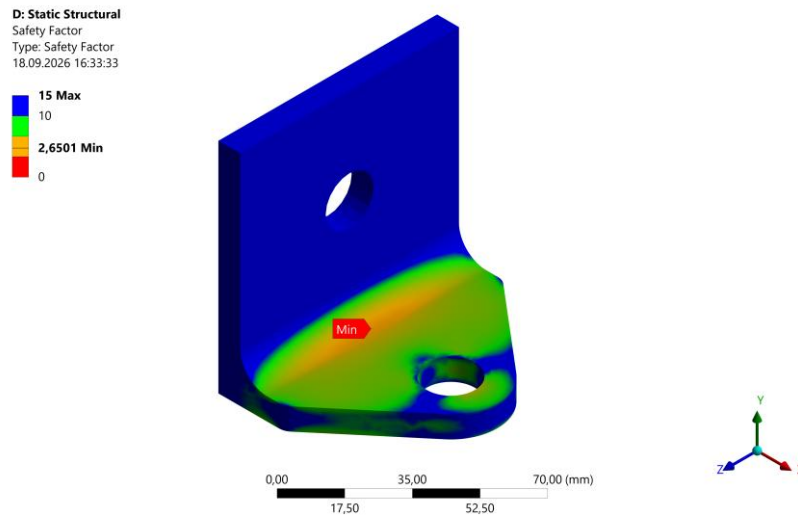


Figure 10.6: Fatigue safety factor distribution of the optimized bracket model

10.7. Fatigue Analysis Results

A fatigue assessment was conducted under fully reversed cyclic loading using the stress-life method, as shown in Figure 10.6. The minimum fatigue safety factor was determined to be 2.65 at the inner radius between the vertical and horizontal sections. As a consequence of this low stress amplitude, the entire structure achieves infinite fatigue life exceeding 10^9 cycles.

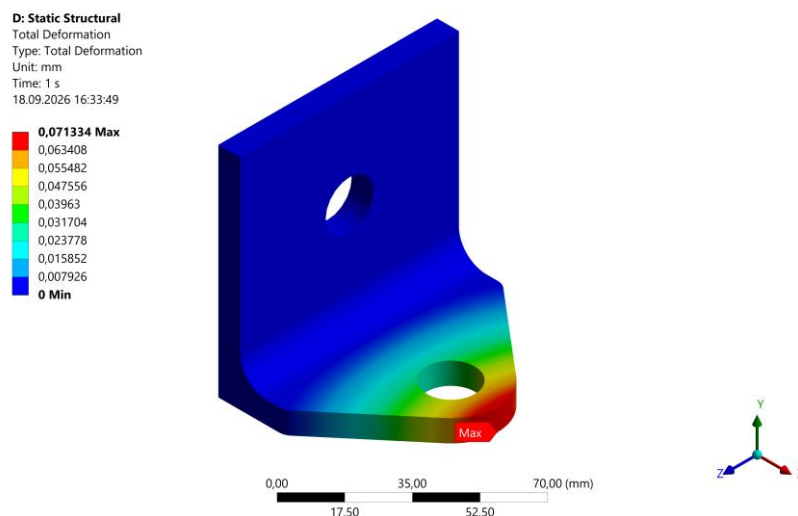


Figure 10.7: Total deformation distribution of the optimized bracket model.

10.8. Deformation Results

The total displacement distribution resulting from the 500 N peak load is presented in Figure 10.7. The maximum deflection is 0.071 mm, located at the free outer edge of the horizontal flange. Considering the 80 mm total cantilever length from the fixed support, this deformation corresponds to approximately $L/1120$, remaining within the standard industrial machine stiffness limit of $L/1000$ (0.080 mm). This confirms adequate structural rigidity and ensures that elastic deflections will not impair functional clearances.

Table 10.2: Comprehensive structural and mass comparison across design iterations.

Metric / Parameter	Baseline Steel Bracket (S235JR)	Aluminum Bracket (EN AW-6061-T6)	Optimized Aluminum Bracket (EN AW-6061-T6)
Material Density [kg/m ³]	7850	2700	2700
Part Mass [g]	742,16	256,90	199,06
Mass Reduction [%]	0%	65,38%	73,18%
Max. Equivalent Stress [MPa]	20,67	20,65	22,64
Yield Strength [MPa]	235	240	240
Static Safety Factor	11,37	11,62	10,60
Max. Total Deformation [mm]	0,027	0,081	0,071
Min. Fatigue Safety Factor	7,02	2,91	2,65

10.9. Comparative Discussion and Technoeconomic Evaluation

The transition from structural steel to the optimized aluminum bracket achieves an overall mass reduction of 73,18%, decreasing the weight from 742,16 g to 199,06 g. Material substitution yields an initial 65,38% saving, while trimming the inactive corner regions provides an additional 22,51% reduction. Under the 500 N operational load, structural integrity is fully maintained: the peak von Mises stress of 22,64 MPa remains well below the 240 MPa yield strength of EN AW-6061-T6 (safety factor: 10.60), and the maximum deflection of 0,071 mm stays strictly within the 0,080 mm ($L/1000$) stiffness limit.

Economically, converting the topology optimization distribution into a standard trimmed geometry ensures low cost fabrication via conventional machining, eliminating the need for expensive additive manufacturing. Although aluminum has a higher base price per kilogram than structural steel, the substantial weight reduction offsets raw material costs while minimizing structural inertia, achieving a practical and cost effective lightweight design.

11. Conclusion

In this study, a structural steel (S235JR) L bracket was redesigned and lightened through material substitution and topology driven geometric optimization under an operational peak load of 500 N. Transitioning to EN AW-6061-T6 aluminum combined with trimming inactive lateral corner material achieved an overall mass reduction of 73.18% (from 742.16 g to 199.06 g). Finite element validation with adaptive mesh convergence confirmed that structural integrity and functional requirements are fully satisfied: the peak von Mises stress of 22.64 MPa yields a static safety factor of 10,60, the fatigue safety factor of 2,65 ensures infinite life exceeding 10^9 cycles, and the maximum deflection of 0,071 mm remains strictly within the standard $L/1000$ (0.080 mm) machine stiffness limit. By replacing organic topology contours with clean planar cuts, the final geometry guarantees low cost manufacturability via standard 3 axis CNC machining, achieving an optimal balance between structural rigidity, weight efficiency, and production economics.

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Appendix A: Technical Drawing of the Baseline L Bracket

